

MARNITZ, Germany

WMO: 102640

Lat: 53.3222N

Lon: 11.9319E

Elev: 87

StdP: 100.28

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.5	-10.5	-16.5	0.9	-12.1	-14.0	1.1	-10.1	11.0	6.0	9.7	4.9	1.6	70	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.4	18.9	26.6	18.0	24.7	17.3	19.9	26.1	18.9	24.9	17.9	23.3	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	22.1	16.6	11.9	21.2	15.6	11.2	20.4	57.2	26.2	53.8	24.7	50.7	23.2	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.1	7.2	-14.0	31.5	6.3	1.3	-18.5	32.5	-22.2	33.3	-25.8	34.0	-30.4	35.0		
			-14.1	22.0	6.0	0.9	-18.5	22.6	-22.0	23.2	-25.3	23.7	-29.7	24.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	0.3	0.0	4.0	7.6	12.3	15.0	17.3	17.3	13.5	9.3	4.3	1.9
	DBStd	7.26	5.85	5.01	3.90	3.60	3.60	3.30	2.95	2.89	2.72	3.59	3.44	3.90
	HDD10.0	1358	302	280	188	90	18	1	0	0	3	55	171	251
	HDD18.3	3606	560	513	444	323	188	112	57	55	146	279	421	509
	CDD10.0	857	0	1	2	18	91	150	225	227	108	34	1	0
	CDD18.3	63	0	0	0	0	2	11	23	24	2	1	0	0
	CDH23.3	691	0	0	0	2	50	136	270	211	22	1	0	0
	CDH26.7	142	0	0	0	0	3	33	59	44	2	0	0	0
Wind	WSAvg	3.3	4.1	3.8	3.7	3.3	2.9	2.8	2.8	2.6	3.0	3.2	3.7	4.0
Precipitation	PrecAvg	620	52	42	41	34	50	63	70	66	51	47	49	54
	PrecMax	848	105	105	96	74	118	119	161	134	147	112	103	113
	PrecMin	446	1	8	7	2	11	14	17	25	12	21	4	12
	PrecStd	101	27	20	22	18	26	27	36	28	30	22	24	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	14.0	16.7	22.0	27.0	30.1	30.3	30.2	25.9	21.4	13.0	11.6
		MCWB	9.1	10.7	10.8	13.7	16.3	19.1	19.6	20.2	17.6	16.3	10.8	9.7
	2%	DB	9.2	10.7	13.4	18.7	24.8	26.9	28.4	27.8	23.2	18.2	11.6	10.4
		MCWB	7.7	8.3	9.3	11.7	15.8	17.6	19.0	18.7	17.2	14.3	10.1	9.0
	5%	DB	8.1	8.1	10.9	16.2	22.3	24.4	26.7	25.8	20.5	16.2	10.3	9.1
		MCWB	6.6	6.2	7.5	10.5	14.3	16.9	18.4	18.0	15.5	12.9	9.0	7.9
	10%	DB	6.8	6.3	9.2	13.9	19.9	22.0	24.2	23.5	18.5	14.5	9.1	7.7
		MCWB	5.4	4.7	7.0	9.3	13.5	15.6	17.6	17.2	14.5	11.8	8.0	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.8	11.4	14.2	18.0	21.0	21.2	21.1	19.1	16.7	11.3	10.2
		MCDB	10.6	13.5	16.0	20.8	24.9	27.2	27.4	28.5	24.1	20.5	12.4	11.1
	2%	WB	8.0	8.4	9.8	12.3	16.4	18.9	20.0	19.5	17.7	14.8	10.2	9.2
		MCDB	9.1	10.5	12.5	17.5	22.7	24.9	26.5	25.5	21.8	18.0	11.3	10.3
	5%	WB	6.7	6.4	8.4	11.0	15.0	17.5	18.9	18.6	16.3	13.2	9.2	7.9
		MCDB	8.0	8.0	10.4	15.3	21.0	23.2	25.3	24.1	19.9	15.4	10.1	8.9
	10%	WB	5.6	4.7	7.1	9.8	13.8	16.2	17.8	17.7	15.0	12.1	8.0	6.6
		MCDB	6.7	6.0	9.1	13.5	19.1	21.2	23.4	22.4	17.8	14.2	8.9	7.6
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	6.5	8.6	10.4	9.5	9.9	9.6	8.2	6.9	4.6	4.0
		MCDBR	4.9	6.6	9.3	12.3	15.0	14.5	14.4	13.5	12.0	9.3	5.8	4.8
		MCWBR	4.5	5.3	6.3	7.0	7.7	7.3	7.0	6.5	7.0	6.0	4.9	4.6
	5% WB	MCDBR	4.6	6.5	8.3	11.4	13.2	12.8	12.9	11.3	10.6	8.3	5.5	4.6
		MCWBR	4.5	5.5	6.0	6.8	7.2	6.9	6.9	6.0	6.8	5.9	5.0	4.5
Clear-Sky Solar Irradiance	taub	0.314	0.336	0.371	0.407	0.403	0.401	0.420	0.414	0.380	0.358	0.334	0.307	
	taud	2.329	2.323	2.258	2.213	2.269	2.295	2.259	2.288	2.363	2.399	2.378	2.326	
	Ebn at Noon	655	758	803	820	846	850	824	806	789	725	626	587	
	Edh at Noon	60	81	106	126	125	123	126	116	95	75	57	50	
All-Sky Solar Radiation	RadAvg	0.59	1.27	2.43	4.02	5.03	5.31	4.98	4.18	3.06	1.68	0.73	0.43	
	RadStd	0.07	0.18	0.26	0.55	0.62	0.41	0.70	0.33	0.34	0.21	0.08	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page